

SHOTLIN

DIGITAL PRODUCT STUDIO



LNDRY Phase 1 MVP - Laundry Service Platform Requirements and Technical Annexure

Service scope, third-party services, security, maps, payment, watermark and acceptance terms

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Source baseline	Approved 34-page workflow and Bakaloo repositories

CONFIDENTIAL - IMPLEMENTATION ANNEXURE

This document is a target implementation contract. It does not state that every listed endpoint already exists in the Bakaloo codebase.

Document status and hierarchy

This document is designed to be attached to the project agreement as the technical and functional annexure. It consolidates the approved workflow into implementable requirements. Where this annexure conflicts with an informal message or verbal statement, the signed agreement and signed annexure control.

Scope lock

Phase 1 includes the customer application, a role-based Vendor Partner application containing Vendor Owner/Staff and Delivery Partner modes, and the Admin Operations Dashboard. Continuous live rider tracking is explicitly excluded.

Service-only commercial model

LNDRY is a laundry-service booking and fulfilment platform. Shirts, trousers, sarees, blankets, shoes and weight-based clothes are customer-owned articles handed over for cleaning or care. They function as garment-rate lines used to calculate the service charge by piece, pair or kilogram.

Service model interpretation

- The customer books a vendor-defined laundry service and selects the garment types, quantities or estimated weight to be processed.
- The amount displayed before pickup is an estimated service charge based on published vendor rates.
- After pickup, the vendor may confirm the actual garment count or measured weight. Any permitted final adjustment uses the same published service rates and is fully audited.
- Pickup and delivery refer to collection of the customer's garments and return of those garments after the selected laundry service is completed.
- The Phase 1 architecture must remain centred on service categories, vendor services, garment rates, pickup-slot capacity, service quotations, laundry processing and return delivery.

1. Service-platform objective

LNDRY is a multi-vendor laundry marketplace where customers discover eligible nearby laundry vendors, compare services and prices, select a vendor, schedule a 60-minute pickup slot, pay through Razorpay, and track status from vendor confirmation through pickup, processing and OTP-verified delivery.

2. Platform surfaces and roles

Surface	Role	Primary responsibility
Customer Mobile App	Customer	Discover, compare, book, pay, track, verify pickup/delivery and review.
Vendor Partner App	Vendor Owner / permitted Vendor Staff	Onboard business, publish services, set prices/capacity, accept orders, assign employees, process orders and view performance.
Vendor Partner App - Rider Mode	Delivery Employee / Rider	View only assigned jobs, open external Google Maps, verify pickup/delivery OTP and complete assigned handovers.
Admin Operations Dashboard	Platform Admin / Finance / Super Admin	Approve vendors, control radius/categories, manage orders/exceptions, reports, security, watermark and platform settings.

3. Functional requirements by module

3.1 Authentication and account security

- Customer, vendor and rider login through mobile OTP.
- OTP challenge expiry, retry limit, resend cooldown and rate limiting.
- Role-scoped sessions when one phone belongs to more than one role.
- Admin dashboard login uses email/password plus mandatory second factor.
- Second factor supports authenticator TOTP; direct email/SMS OTP may be configured as fallback.
- Recovery codes, session revocation, forced password change and audit logs for admin security.

3.2 Vendor application and approval

- Owner details, business details, operating hours, working days, pickup/delivery availability and optional GST.
- Owner identity, shop photograph, registration document, GST certificate and configurable document types.
- Exact business pin with latitude and longitude.
- Requested radius is not final; admin approves the effective radius.
- Statuses: DRAFT, WAITING_FOR_APPROVAL, CORRECTION_REQUIRED, APPROVED, REJECTED, SUSPENDED.
- Approval, rejection and correction notifications by enabled channels.

3.3 Vendor public profile and service catalogue

- Public laundry name, logo/shop image, description, contact, hours, working days and pickup/delivery capability.
- Admin-managed categories such as Wash & Fold, Wash & Iron, Dry Cleaning, Ironing, Shoe Care, Blanket, Curtain and Carpet Cleaning.
- Vendor-defined service name, description, inclusions, exclusions, completion time and optional image.
- Clothing/garment types priced per piece, kilogram or pair with minimum and maximum quantity.
- A profile becomes discoverable only after vendor approval, activation, effective radius and at least one published valid service.

3.4 Capacity and pickup slots

- Vendor defines daily order limit, active days, 60-minute slots and maximum orders per slot.
- A vendor without configured capacity is unavailable.
- Only slots with remaining capacity are shown to customers.
- Booking confirmation creates a short-lived slot hold; payment failure, timeout or rejection releases capacity.
- Database-level atomic enforcement prevents two customers booking the final slot.
- Admin may inspect and apply audited temporary overrides.

3.5 Customer discovery and comparison

- Customer chooses current location, searches address or selects a saved address, then confirms an exact pin.
- Eligibility checks customer coordinates, vendor coordinates, approved radius, vendor status, service status, capacity and slot availability.
- Nearest eligible vendors appear first by default; the customer makes the final selection.
- Sorting: nearest, price low-to-high, price high-to-low, best rating and value for money.
- Filtering: available now and service/garment type.
- Search supports service name, category name, garment type and vendor name.

3.6 Quotation, order and payment

- Customer selects a laundry service, garment quantities or estimated weight, slot and address.
- Server calculates estimate using a service-rate snapshot; the mobile client cannot be source of truth for amount.
- Razorpay supports enabled UPI, card, net banking and wallet methods.
- After verified payment, order is sent only to the customer-selected vendor.
- Initial business status is Waiting for Vendor Confirmation.
- Vendor accepts or rejects; no response triggers configurable auto-reject.
- The system does not automatically assign another vendor after rejection.

3.7 Pickup, processing and delivery

- Accepted order receives an available pickup employee assignment; vendor may reassign.
- Rider opens exact customer location in external Google Maps and marks Going for Pickup.
- Pickup requires purpose-bound pickup OTP before completion.
- Vendor confirms Received at Partner and may confirm actual garment count or measured weight.
- Processing stages support Received, Washing, Drying, Ironing and Packed; irrelevant stages may be skipped through permitted transitions.
- Packed order receives delivery assignment; rider marks Out for Delivery.
- Delivery requires separate purpose-bound delivery OTP before Delivered.

3.8 Tracking, history, ratings and reports

- Customer sees status timeline and notifications, not a continuously moving rider marker.
- Order history includes vendor, garment lines, quantities/weight, timestamps, payment, final amount and status history.
- Only delivered orders are review-eligible.
- Vendor performance includes acceptance, completion, average delivery time, rating and counts.
- Vendor revenue includes daily, weekly, monthly and per-order details.
- Admin reports include order volume, GMV, customer activity, vendor performance, completion and cancellation.

4. Map and location implementation

Use case	Required technology	Behaviour
Customer mobile address picker	Google Maps SDK for Flutter plus Google Places Autocomplete and Geocoding.	Search, current location, draggable exact pin and saved lat/lng.
Vendor business pin	Same Google Places and pin-selection flow.	Stores address, place ID, latitude and longitude.
Admin map preview	Reuse React-Leaflet/Leaflet in Bakaloo dashboard or approved Google Maps web component.	Read-only verification of stored coordinates and approved radius.
Distance/radius eligibility	Backend Haversine calculation using PostgreSQL query or PostGIS if introduced.	Avoids a paid route API for straight-line service-radius decisions.
Rider navigation	External Google Maps deep link / URL intent.	Google Maps handles route, travel time, traffic and turn-by-turn directions.
Customer tracking	No embedded live navigation map and no continuous rider GPS feed.	Status timeline and push notifications only.

Map billing responsibility

Google Maps Platform keys, billing account, quotas and any per-request charges must be provided and paid by the client unless the commercial agreement explicitly includes them.

5. Third-party services and infrastructure

Service	Purpose	LNDRY use	Status
PostgreSQL	Primary transactional database	Users, vendors, services, service rates, slots, orders, payments, events and audit logs.	Required
Redis	Cache and short-lived state	OTP challenges, rate limits, slot holds, idempotency and BullMQ coordination.	Required

Service	Purpose	LNDRY use	Status
BullMQ	Background queue	Auto-reject, slot expiry, notifications, watermark jobs and retryable work.	Required
Razorpay	Payment gateway	Order creation, signature verification, webhooks and refunds.	Required
Firebase Cloud Messaging	Push notifications	Customer, vendor and rider operational notifications.	Required
Firebase Crashlytics	Mobile diagnostics	Crash reports for Flutter applications.	Recommended
Firebase Analytics	Service-platform usage analytics	Screen and funnel events without storing sensitive OTP/payment values.	Recommended
Cloudinary	Image/document asset processing	Public media, transformations and signed/restricted document access.	Required or equivalent
Google Maps Platform	Places, map pin, geocoding	Address search and exact coordinates.	Required
Google Maps external navigation	Rider route app	Deep-link to customer/vendor coordinates.	Required
SMS provider - Twilio or India DLT-compliant provider	Mobile OTP and transactional SMS	Login OTP and selected operational notifications.	Required
Email provider - Resend/SMTP	Email notifications	Vendor approval, rejection, correction and administrative messages.	Recommended
Socket.IO	Real-time application events	Order, assignment and notification updates; FCM remains fallback.	Recommended
Cloud/VPS and reverse proxy	Runtime hosting	Node/Fastify API, worker, PostgreSQL/managed DB, Redis, HTTPS and backups.	Required

6. Watermark and secure-document requirements

- Admin Dashboard includes a configurable watermark module with logo or text, position, scale, opacity and application profile.
- Marketing watermark may be applied to vendor service/shop images as a generated derivative while retaining the original according to policy.
- Sensitive KYC previews must display a dynamic visible watermark such as "For LNDRY Verification Only", vendor/application reference and view date.
- Original KYC files remain private; access uses short-lived signed URLs or authenticated streaming with audit logging.
- Watermark processing is server-side using Sharp/Cloudinary transformations and BullMQ for batch processing.
- Admin can preview, enable/disable, apply, retry failed jobs and inspect watermark history.
- Watermark is not a substitute for encryption, access control or document retention policy.

7. OTP and two-factor requirements

OTP type	Use	Mandatory control
Login OTP	Customer/Vendor/Rider login	Mobile-number challenge; hashed, expiring, rate-limited and device/session linked.
Pickup OTP	Customer to assigned pickup rider	Order-specific, PICKUP purpose, unavailable before assignment, single-use.

OTP type	Use	Mandatory control
Delivery OTP	Customer to assigned delivery rider	Order-specific, DELIVERY purpose, single-use and independent from pickup OTP.
Admin second factor	Dashboard security	TOTP authenticator preferred; direct email/SMS OTP fallback if enabled.
Step-up OTP/2FA	Refund, admin override, security changes	Recent second-factor proof required for high-risk action.

8. Data model requirements

- Core entities: users, roles, devices, addresses, vendors, vendor applications, documents, employees, categories, services, garment types, prices, recurring slots, slot exceptions, slot holds, orders, order garment lines, assignments, OTP challenges, payments, refunds, reviews, notifications, audit logs, watermark settings and asset jobs.
- Orders store immutable snapshots of vendor identity, address, service, service rate, selected garment quantity/estimated weight, slot and estimated amount.
- Price changes after booking do not retroactively change existing order snapshots.
- Final quantity/weight adjustments retain original and confirmed values plus reason and actor.
- Soft deletion is used for records that must remain referenced by historical orders.
- Every critical state change writes an order event with actor, old state, new state, timestamp and request ID.

9. Security and non-functional requirements

Area	Requirement
Transport	HTTPS only in production; secure headers and trusted proxy configuration.
Authentication	Short-lived JWT access tokens, rotating refresh tokens, session revocation and role scope.
Authorization	Backend-enforced RBAC/permission strings and tenant scoping on every protected endpoint.
Input safety	AJV/Zod validation, multipart file validation, sanitisation and safe error responses.
Rate limiting	OTP, login, payment, upload and high-risk endpoints have tighter limits.
Secrets	Environment/secret manager only; never committed to GitHub or mobile source.
Payments	Amounts computed server-side; webhook signature and idempotency required.
PII	Least-privilege access, masked logs, audited document views and retention/deletion policy.
Availability	Health/readiness checks, graceful shutdown, database backups and worker retry strategy.
Observability	Structured logs with request IDs, error monitoring and operational dashboards.
Performance	Pagination, indexes, cached public data and no unbounded queries.
Accessibility	Readable contrast, scalable text, labelled controls and keyboard-accessible dashboard.

10. Explicit Phase 1 exclusions

- Continuous live rider movement on the Customer App.
- Automatic reassignment to another vendor after rejection or timeout.
- Dynamic surge pricing, subscription plans, loyalty programmes or unrelated marketplace modules unless separately approved.
- In-app turn-by-turn navigation; rider uses external Google Maps.
- Automated government KYC verification unless a separate provider and commercial scope are approved.
- App Store/Play Store account fees, Google Maps charges, SMS charges, Razorpay charges, Cloudinary charges and cloud-hosting fees unless explicitly included in the commercial agreement.

11. Client-provided dependencies

- Approved brand assets, legal business name, privacy policy, terms, refund/cancellation policy and support contacts.
- Razorpay account and production keys with webhook access.
- Google Cloud project, Maps billing, API keys and package/domain restrictions.
- Firebase project access and platform configuration files.
- Cloudinary account or approved alternative.
- SMS/DLT account, approved templates and sender/entity details for India.
- Production domain, server/VPS or cloud account and DNS access.
- Final category, service, pricing, cancellation, auto-reject and refund business decisions.

12. Acceptance criteria

Area	Acceptance condition
Vendor onboarding	Vendor submits full application; admin can approve, reject or request correction; effective radius is admin-controlled.
Discovery	Only eligible vendors within approved radius and with active service/capacity appear.
Capacity	Concurrent attempts cannot exceed slot or daily capacity.
Payment	Successful Razorpay signature creates one paid order; duplicate webhook does not duplicate order.
Vendor confirmation	Vendor can accept/reject; configured timeout auto-rejects and notifies customer.
Pickup	Only assigned rider can verify pickup OTP and complete pickup.
Processing	Vendor can use permitted laundry states; customer sees grouped timeline.
Delivery	Only assigned rider can verify delivery OTP and mark delivered.
Security	Admin 2FA is enforced; high-risk actions are audited and step-up protected.
Watermark	Configured watermark preview and derivative generation work; private documents remain access-controlled.
Reports	Vendor and admin metrics match underlying delivered/rejected/cancelled orders.

13. Change-control rule

Any new feature not stated in this annexure is a change request. The change must record scope, API/database impact, user-interface impact, third-party cost, delivery impact and commercial approval before development begins.